

Waterfall

A sequential model organizing work into phases with strong emphasis on up-front definition and planning.

What is Waterfall?

Waterfall organizes development into sequential phases, commonly requirements, analysis/design, development, testing, deployment and maintenance. Each phase feeds the next; late changes may require revisiting earlier decisions and artifacts.

Where QA participates

Even with a formal test phase after development, QA can contribute early by reviewing requirements, ambiguity, rules, testability, risks, data, environments and strategy. Early QA reduces the chance of discovering fundamental issues only at the end.

Traceability and RTM

In projects with formal requirements, a Requirements Traceability Matrix can connect requirement, test case, execution and defect, helping answer which requirements are covered and impacted by change.

Practical example

Requirement: maximum transfer of \$10,000. QA asks whether exactly \$10,000 is accepted, \$10,000.01 rejected, whether daily limits exist and which message appears. Scenarios can be prepared before formal execution.

Change and impact analysis

When requirements change after design or implementation, impact can affect code, documentation, tests, data and integrations. Change control and impact analysis reduce inconsistencies.

Waterfall and agile approaches

Waterfall emphasizes up-front definition and sequential phases; agile approaches favor short cycles, frequent feedback and adaptation. Real projects may use hybrid models. Context determines useful practices.

Watch out

A risk of highly sequential flow is discovering defects or wrong requirements late, when rework costs more. Reviewing requirements and testability early remains valuable.